BT301 – Assignment 01,Fall 24

Due Date: 22-11-2024

Assignment total marks: 15

Assignment Tasks

Read the Research Paper
Study the following paper to understand structural and regulatory differences between prokaryotic and eukaryotic genes:

> Comparative analysis of gene structures and expression patterns in prokaryotes and eukaryotes.
> Link: https://pmc.ncbi.nlm.nih.gov/articles/PMC6137564/

1. **Summarize the Key Differences (100-150 words)**
   Based on the paper, summarize the main structural and regulatory differences between prokaryotic and eukaryotic genes in your own words. Focus on aspects like operons, introns, and gene regulation mechanisms.
2. **Practical Implications (50-100 words)**
   Explain why understanding these differences is important in biotechnology, particularly in applications like gene editing, synthetic biology, or protein production.
3. **Citation**
   Write the correct citation for this paper in APA format.

Formatting: Length: 200-300 words. Use Times New Roman, 12-point font, and double spacing.

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Solution Starts here 👇

1.Summary of Key Differences (100-150 words)

Prokaryotic genes are organized into operons, allowing multiple genes to be transcribed as a single mRNA, whereas eukaryotic genes are transcribed individually. Eukaryotic genes contain introns, which are spliced out post-transcription, whereas prokaryotic genes lack introns. Regulation in prokaryotes is primarily controlled at the transcriptional level through simple promoter and operator interactions, often involving repressor or activator proteins. In contrast, eukaryotic regulation involves complex interactions, including chromatin remodeling, enhancer-promoter looping, and the involvement of numerous transcription factors. Eukaryotic genes also undergo extensive RNA processing, such as capping and polyadenylation, which are absent in prokaryotes. These structural and regulatory differences reflect the greater cellular and functional complexity of eukaryotes.

2. Practical Implications (50-100 words)

Understanding the differences between prokaryotic and eukaryotic genes is crucial in biotechnology. It enables precise manipulation in gene editing and synthetic biology, such as designing expression systems suited to the target organism. For example, eukaryotic systems are essential for producing complex proteins requiring post-translational modifications. Similarly, bacterial systems, with simpler regulation, are preferred for high-yield protein production in pharmaceuticals.

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References (APA)

Chowdhury, N., & Bagchi, A. (2018). Comparative analysis of prokaryotic and eukaryotic transcription factors using machine-learning techniques. *Bioinformation, 14*(6), 315-326.

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